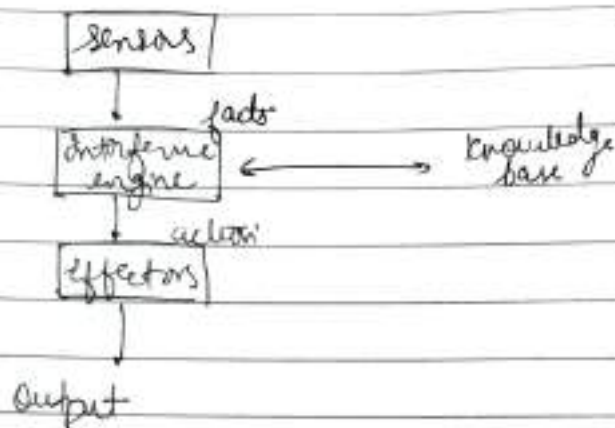


Assignment 2

1) What is Knowledge Based Agent?
Environment

Input

agent



- A knowledge-based system is a system that uses artificial intelligence techniques to store & reason with knowledge. The knowledge is typically represented as in the form of rules or facts, which can be used to draw conclusions or make decisions.

- One of the key benefits of a knowledge based system is that it can help to automate decision making process.

- Another benefit is that they can be used to explain their decisions to humans.

- Knowledge based systems are a type of AI & have been used in variety of applications including medical diagnosis, expert systems & decision support systems.

2] Explain the various methods of knowledge representation techniques.

→ Methods of Knowledge Representation

1. Logic based Systems

• Logic based methods use formal rules to model knowledge. These systems prioritize precision & are ideal for deterministic environments.

• Propositional Logic :- Represents knowledge as declarative statements linked by logical operators like AND, OR, & NOT.

• First-order Logic (FOL) :- Extends propositional logic by introducing variables, quantifiers & predicates. FOL can express statements like "All humans ($\forall x$) are mortal (Mortal(x)). It supports nuanced reasoning but demands significant computational resources.

2. Structured Representations

These methods organize knowledge hierarchically or through networks.

• Semantic Networks :- Represents knowledge as nodes & edges.

• Frames :- Group related attributes into structured "frames".

• Ontologies :- Define concepts, hierarchies, & relationships within a domain using standard like OWL. Ontologies power semantic search engines & healthcare diagnostics by standardizing terminology.

3. Probabilistic Models

These systems handle uncertainty by assigning probabilities to outcomes.

- Bayesian Networks.

Use directed graphs to model causal relationships. Each node represents a variable, & edges conditional dependencies.

- Markov Decision Process (MDPs)

Model sequential decision-making in dynamic environments. MDP help robotic systems navigate obstacles by evaluating potential actions & rewards.

4. Distributed Representation

Modern AI leverages neural networks to encode knowledge as numerical vectors, capturing latent patterns in data.

- Embeddings: convert words, images or entities into dense vectors. Word embeddings like word2vec map synonyms to nearby vectors enabling semantic analysis.

3] write short note on predicate logic:

→ Predicate logic is also known as First Order Logic (FOL) is a formal system used in mathematical logic, & AI to express statements with quantifiers, variables & predicates.

Key components of Predicate Logic

1. Constants - Represents specific objects (eg. John, 5).
2. Variables - Represents general entities (eg. x, y).
3. Predicates - Describe properties or relationships between objects (eg. $\text{parent}(\text{John}, \text{Mary})$).
4. Quantifiers:
 - Universal Quantifier ($\forall$) - Represents "for all".
 - Existential Quantifier ($\exists$) - Represents "there exists".

5. Logical Connectives - Includes AND ($\wedge$), OR ($\vee$), NOT ($\neg$) IMPLIES ($\rightarrow$).

- Applications:

- Artificial Intelligence (AI) (Knowledge representation, Expert Systems).
- Database Query language
- Automated Theorem Proving

4] Explain forward chaining & backward chaining algorithm with example

→ It is fundamental reasoning methods used in rule-based systems & AI. The primary difference lies in their approach to problem solving.

• Forward Chaining:-

forward chaining is fact-based starts with available facts & applies rules to derive new information moving step by step toward conclusion. In desired outcome & working backward to identify the facts & rules needed to support it effectively for targeted problem-solving, such as diagnosis diseases or troubleshooting issues. Forward chaining is ideal for scenarios where real-time data is abundant, such as monitoring systems or diagnosis.

eg. of forward chaining

1. Fire Alarm Systems.
2. Recommendation System.
3. Traffic Light Control
4. Smart Home System

eg of Backward Chaining

Backward Chaining is a top down approach that evaluates whether the goal can be reached by verifying the necessary facts & rules. It prioritizes efficiency & relevance, making it ideal for solving specific problems with a clear goal in mind.

1. Medical Diagnosis
2. Troubleshooting a Car Engine
3. Customer Support Chatbots
4. Educational Tutoring Systems

5) Why Uncertainty occurs in the AI systems, How it apply probability, toothache problem

→ Uncertainty in AI arise due to incompleteness, ambiguity or randomness of data & knowledge. The main causes include:

1. Lack of complete information
2. Noisy or inaccurate data
3. Ambiguity in language
4. Probabilistic events

Applying probability to the Toothache Problem

We use ~~basic~~ Bayesian Probability to model uncertainty in AI. The toothache problem can be analyzed using conditional probability.

Problem definition:

Let,

- Cavity (C) be the event of having cavity
- Toothache (T) be the event of experiencing toothache

- $P(C)$ = Probability of having cavity
- $P(T|C)$ = Probability of toothache given a cavity
- $P(C|T)$ = Probability of having given a toothache

Using Bayes Theorem:

$$P(C|T) = \frac{P(T|C) \cdot P(C) \cdot P(C|T)}{P(T)} = \frac{P(T|C) \cdot P(C)}{P(T)}$$

where $P(T)$ is the overall probability of toothache

6] Write short note on belief networks

→ A Belief network also known as a Bayesian Network (BN) is a graphical model used to represent probabilistic relationships among variables in an AI system. It helps in reasoning under uncertainty by using Bayes theorem.

Key Components of Belief Network:

1. Nodes: Represent random variables (e.g. Disease, Symptoms)
2. Edges: Directed links showing dependencies between variables

3. Conditional Probability Table (CPT): Define the probability of each variable to its parents.

- Applications:

- Medical Diagnosis
- Speech & Image Recognition
- Robotics & AI Decision Making

Unit-5

1) What is planning in AI?

→ Planning refers to the process of determining a sequence of actions that a system must take to achieve a specific goal. Just like humans plan their daily tasks with a goal in mind, AI systems use planning algorithms to break down a problem into smaller, manageable tasks & determine the best course of action.

Planning is crucial for AI applications because it allows machines to

- Think Ahead! - AI systems evaluate multiple possible outcomes before selecting the optimal one.
- Adapt to changes - when faced with uncertain environments, AI systems can revise their plans based on new information.
- Act automatically - whether it's an autonomous robot or a smart assistant, AI planning enables machines to perform tasks independently.

2) Compare & contrast problem solving agent with planning agent

Feature	Problem-Solving Agent	Planning Agent
Definition	Solves problems by searching for a sequence of actions to reach a goal.	Uses a structured approach to plan a sequence of actions before execution.
Approach	Uses search algorithms (e.g. BFS, DFS, A*)	Uses planning techniques (STRIPS, PDDL).

Knowledge Representation	Minimal, often words with state-space representation	Uses explicit knowledge representation
Execution	Generates a solution & execute it step by step	Plans a complete sequence of actions before execution
Flexibility	May need re-planning if the environment changes	Can modify plans dynamically based on conditions
Efficiency	Can be computationally expensive due to blind search	More efficient as it considers future actions in advance
eg.	A may solve by searching for the exit	A robot can plan a sequence to assemble a product

3] Explain partial order planning, hierarchical planning & conditional planning

→ 1. Partial-Order planning (POP)

- A non-linear planning approach where actions are partially ordered rather than strictly sequential
- The planner focuses on achieving sub-goals independently before fully ordering action.
- Uses causal links to ensure dependencies between actions are maintained.

eg. In robotic assembly, a robot can pickup screws & fetch a screwdriver in any order, but must tighten the screw after picking both the items.

2. Hierarchical Planning (HTN) - Hierarchical Task Network

- Breaks down complex tasks into smaller sub-tasks.
- Uses high level actions that decompose into lower level actions.
- The decomposition continues until reaching primitive actions executable by the agent.
- eg. A robot chef planning to cook a meal.

Advantages:

- More efficient for large problems by forming task hierarchy.
- Reduces the search space by using predefined action hierarchies.

3.

3. Conditional Planning

- Also known as contingency planning, it handles uncertainty by planning for multiple possible outcomes.
- Uses if-then conditions to modify the plan dynamically based on observations.
- Includes sensing actions that gather information before taking further steps.
- eg. A self-driving car planning to turn at an intersection.

4) Explain various methods of learning?

→ 1. Supervised learning

Supervised learning is a type of machine learning where an algorithm is trained on a labeled dataset, meaning that both the input & the correct output are known. It generalizes the patterns to make predictions on new, unseen data.

eg. Image classification

2. Unsupervised learning.

• Unsupervised learning is a type of machine learning where the algorithm is trained on unlabeled data, meaning there are no predefined output. The goal is to discover hidden patterns, relationships, or structures in the dataset.

eg Customer segmentation in marketing

- Types -

• Clustering: Groups similar data points (eg K-means)

• Association: Identifies relationships in data

- Algorithms: K-means, DBSCAN, PCA, Apriori

Applications: Anomaly detection, recommendation system, image segmentation

3. Reinforcement learning

It is a type of ML where an AI agent interacts with an environment & learns by trial & error. The agent receives rewards for correct actions & penalties for incorrect actions, helping it optimize the decision-making strategy.

eg Training an AI agent to play chess.

- Algorithms: Q-learning, SARSA, Deep Q-learning

Applications: Self-driving cars, AI for healthcare optimization, financial trading.

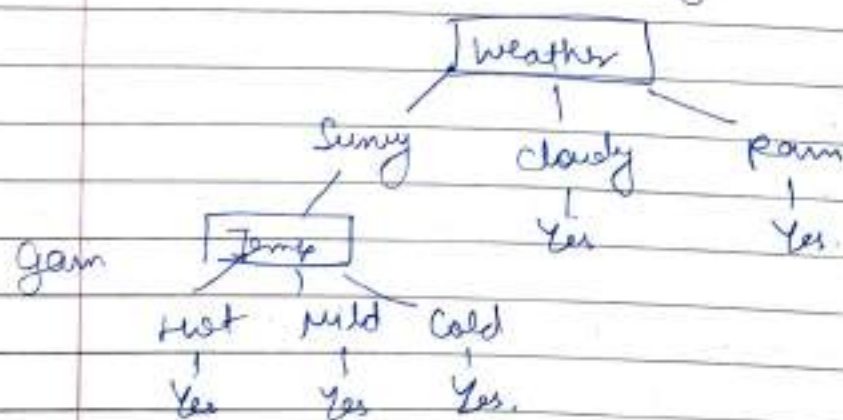
5] Compare supervised, unsupervised, & reinforcement learning with example.

Definition	learns from labeled data	learns from unlabeled data	learns by interacting with an environment
Goal	Predict outcomes based on input	Find hidden patterns or data structure	Maximize rewards through trial & error

Data type	Labelled data	Unlabelled data	Feedback or penalties
eg	Image classification, Spam detection	Customer Segmentation, anomaly detection	Game playing, Robotics, self-driving cars

Q] What is decision tree? Explain decision tree with example

- A decision tree is a flowchart like diagram mapping out all the of the potential solution to a given problem. They're often used by organizations to help determine the most optimal course of action by comparing all of the possible consequences of making a set of decisions.
- Decision trees are also a popular in machine learning, as they can be used to build predictive models. These types of decision trees can be used to make predictions such as whether a customer will buy a product based on their previous purchase history.



Unit - 6

1) Explain different components of NLP.
⇒ Natural Language Processing (NLP) is a branch of Artificial Intelligence (AI) that enables machines to understand, interpret & generate human language. NLP consists of several key components that help process & analyze text or speech data.

1. Lexical Analysis

- Focuses on analyzing the structure of words & breaking text into tokens.
- Identifies meanings, roots & parts of speech.
- e.g. Breaking the sentence.

2. Syntactic Analysis (Parsing)

- Deals with the grammatical structure of sentences.
- Ensure that words are arranged in meaningful way using grammar rules.

3. Semantic Analysis (Meaning Extraction)

- Focuses on understanding the actual meaning of words & sentences.
- Helps resolve ambiguity & interpret context.

4. Pragmatic Analysis (Context Understanding)

- Interprets real-world meaning beyond literal text.
- Considers speaker intent, tone & cultural context.

5. Discourse Analysis (Sentence Relationship)

1. Examines how sentences relate to each other in a conversation or document.
- Helps in text summarization, coherence & topic modelling.

2) Explain the level of knowledge used in NLP.

1. Phonology level
 - It is the lowest level, it deals with pronunciation.
 - It deals with the interpretation of speech across words.
2. Morphological level
 - It deals with the smallest words that convey meaning & suffixes & prefixes.
 - Morphemes mean studying that are built from smaller meanings.
 - E.g. the rabbit has single morphemes while the rabbits have two morphemes.
 - The 's' denotes the singular & plural concepts.
3. Lexical level
 - This deals with the study at the level of words with respect to their lexical meaning & Part of Speech.
 - It uses the lexicon that is collection of lexemes.
 - A lexeme is a basic unit of lexical meaning which is an abstract unit of morphological analysis.
4. Syntactic level
 - This level deals with the grammar & structure of

- It studies the proper relationships between the words.
- The POS tagging output of lexical analysis can be used at the syntactic level of two group words into the phrase & clause brackets.

5. Semantics level :

- This level deals with the meaning of words & sentences
- There are different two approaches:
 - 1) Syntax driven semantic analysis
 - 2) Semantic grammar
- It's a study of meaning of words that are associated with grammatical structure.

6. Discourse level

- This level deals with the structure of different kinds of text.
- There are 2 types of discourse:
 - 1) Anaphora resolution
 - 2) Discourse / text structure recognition
- Words are replaced in anaphora resolution

7. Pragmatic level

This level deals with the real world knowledge & understanding of how this influences the meaning of what is being communicated. Pragmatics identifies the meaning of words & phrases based on how language is used to communicate.

3] Explain Application of NLP in brief.

→ 1. Chatbots

- Chatbots are a form of AI that are programmed to interact with humans in such a way that they sound like humans themselves.
- Depending on the complexity of the chatbots, they can either just respond to specific keywords or they can even hold full conversations that make it tough to distinguish them from humans.

2. Autocomplete in Search Engines

- Search engines use their enormous data sets to analyze what their customers are probably typing when they enter particular words & suggest the most common possibilities. They
- They use NLP to make sense of these words & how they are interconnected to form different sentences.

3. Voice Assistants

- They use a complex combination of speech recognition, natural language understanding, & natural language processing to understand what humans are saying & then act on it. The long term goal of voice assistants is to become a bridge between humans & the internet & provide all manner of services based on just voice interaction.

4. Sentiment Analysis

Almost all the world is on social media these days! And companies can use sentiment analysis to understand how a particular type of user feels about a particular topic, product, etc. They can use natural language processing, Computational Linguistics, text analysis, etc. to understand the general sentiment of the users for their products & services & find out if the sentiment is good or bad, neutral. Companies can use sentiment analysis in a lot of ways such as to find out the emotions of their customers, to understand product reviews, etc.

4) Explain the role of robotics in AI with eg.

- Robotics in AI involves integrating AI technologies into robotic systems to enhance their capabilities & enable them to perform more complex tasks.
- AI in robotics allows robots to learn from experience, adapt to new situations, & make decisions based on the data from sensors. This can involve ML, CV, NLP & other AI techniques.
- AI Robots can use machine learning to analyze data, patterns, & improve performance over time. This is useful for tasks where the environment is dynamic & unpredictable.
- AI-powered vision systems enable robots to interpret & understand visual information from the surroundings. This is crucial for tasks like object recognition, navigation & manipulation.

— Uses in Robotics

1. Computer Vision

— AI Algorithms process camera & sensor data to map surroundings, identify obstacles, & plan safe & efficient paths for robots to navigate.

2. NLP

— Sentiment Analysis can be analyzed by AI using text to speech to understand the emotions & adjust robot behaviour accordingly.